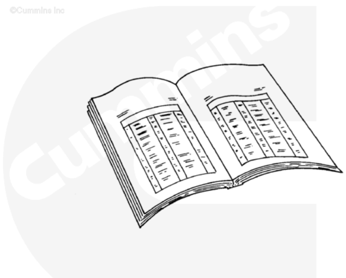
Trainings

Preparatory Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Do not bleed the fuel system of a hot engine; this can result in fuel spilling onto a hot exhaust manifold, which can cause a fire.



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- Remove the lubricating oil filters. Refer to Procedure 007-013 (/qs3/pubsys2/xml/en/procedures/20/20-007-013-tr.html).
- Remove the fuel pump (if the engine is equipped with an air compressor). Refer to Procedure 005-016 (/qs3/pubsys2/xml/en/procedures/20/20-005-016-tr.html).
- Remove the air compressor, if equipped. Refer to Procedure 012-014 (/qs3/pubsys2/xml/en/procedures/20/20-012-014-tr.html).
- Remove the electronic fuel control valve assembly. Refer to Procedure 006-049 (/qs3/pubsys2/xml/en/procedures/20/20-006-049-tr.html).

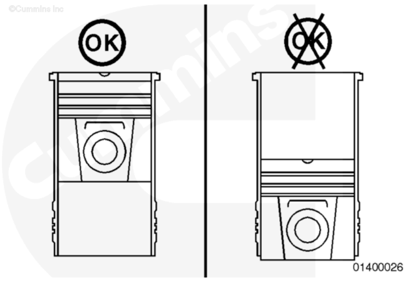
with Electronically Actuated Injector

- Remove the fuel pump. Refer to Procedure 005-016 (/qs3/pubsys2/xml/en/procedures/20/20-005-016-tr.html).
- Remove the ECM cooling plate. Refer to Procedure 006-006 (/qs3/pubsys2/xml/en/procedures/20/20-006-006-tr.html).

Remove

⚠ CAUTION ⚠

Do not attempt to remove the piston cooling nozzle when the piston is at bottom dead center. The nozzle will be damaged and can result in piston failure.

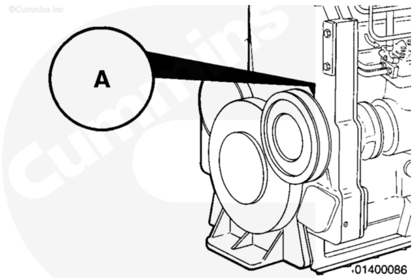


To prevent damage to the piston cooling nozzles during removal, use the valve set marks and the sequence chart to identify those nozzles that can be removed at each index mark on the accessory drive pulley.

One complete crankshaft revolution is required to remove all of the nozzles.

Valve Set Mark	Remove Piston Cooling Nozzles on Cylinders
A	1, 6
B	2, 5
C	3, 4

Rotate the engine barring device until the "A" mark on the pulley is aligned with the mark that is cast on the accessory drive seal boss.

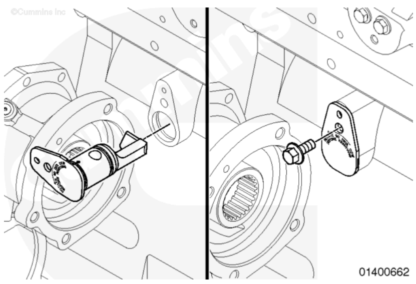


Remove the piston cooling nozzle on cylinders number 1 and number 6.

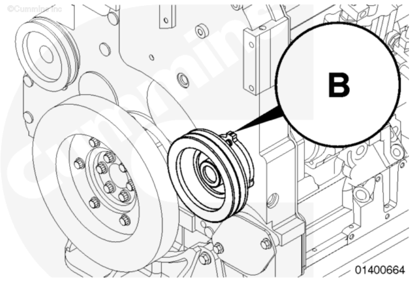
Remove the piston cooling nozzle mounting capscrew.

Turn the nozzle 90 degrees in the direction of the arrow stamped on the nozzle.

Carefully use a pry bar on the nozzle flange to remove the nozzle from the cylinder block.



After removing the nozzle from cylinders number 1 and number 6, rotate the engine barring device and align the next valve set mark with the mark cast on the gear cover.



Use the chart, for reference, remove the appropriate piston cooling nozzles.

Repeat the process to remove the remaining piston cooling nozzles.

Valve Set Mark	**Remove Piston Cooling Nozzles on Cylinders**
A	1, 6
B	2, 5
C	3, 4

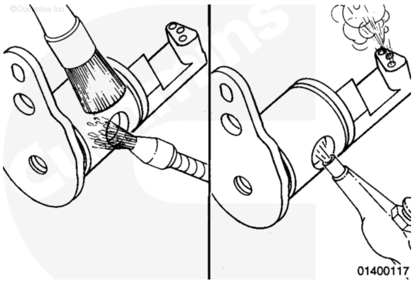
Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

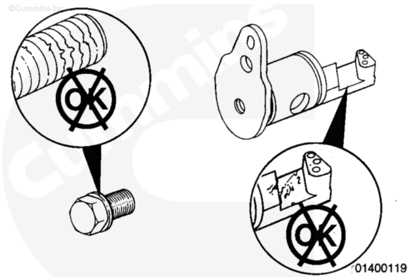


Use solvent, Part Number 3824421, or equivalent, to clean the piston cooling nozzles.

Dry the cooling nozzles with compressed air and blow out the oil passages.

⚠ CAUTION ⚠

Any damage to the piston cooling nozzle can result in engine damage.



Inspect the piston cooling nozzles for cracks or damage.

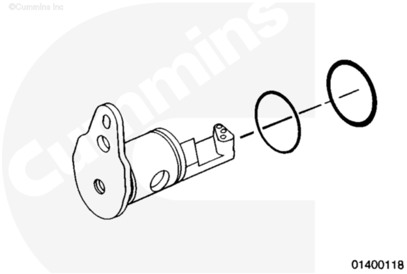
Inspect the screws for damaged or mutilated threads.

Replace any parts that are damaged.

Install

⚠ CAUTION ⚠

Use care when handling the piston cooling nozzle. Any damage to the piston cooling nozzle can result in engine damage.

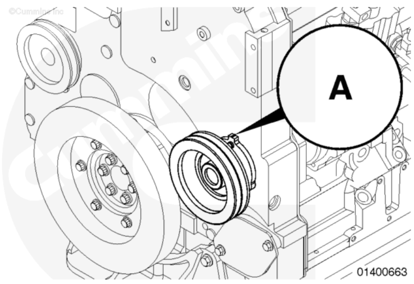


Use vegetable oil to lubricate the new o-ring. Do **not** soak the new o-ring in engine oil.

Install the o-ring in the groove of the piston cooling nozzle that is the least distance to the flange.

Install the piston ring type (steel) seal ring in the groove that is the greatest distance from the flange.

Align one of the valve set marks with the mark cast in the accessory drive seal boss.

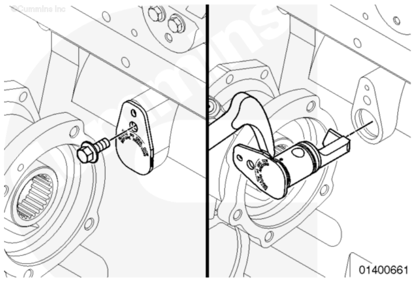


Reference the sequence chart illustrated to identify the two piston cooling nozzles that can be installed.

Valve Set Mark	Install Piston Cooling Nozzles on Cylinders
A	1, 6
B	2, 5
C	3, 4

⚠ CAUTION ⚠

Do not use the capscrew to pull the nozzle into the cylinder block. Nozzles and o-ring damage or external oil leaks can result.



Place the nozzle in the block with the nozzle end parallel to the crankshaft centerline.

Install the two piston cooling nozzles. Use **only** your hand to push the nozzle into the bore until the flange touches the block.

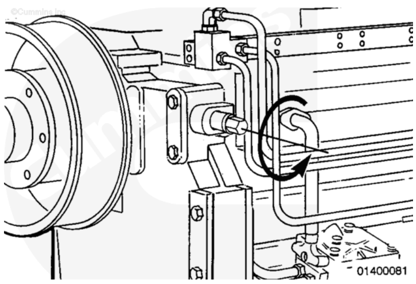
Turn the nozzle 90 degrees to align the capscrew hole.

Install and tighten the capscrew.

Torque Value: 45 n•m [33 ft-lb]

Rotate the barring device and align the next valve set mark with the mark cast into the gear cover.

Repeat the process to install the remaining piston cooling nozzles.



Finishing Steps

with Mechanically Actuated Injector

- Install the electronic fuel control valve assembly. Refer to Procedure 006-049 (/qs3/pubsys2/xml/en/procedures/20/20-006-049-tr.html).
- Install the air compressor if equipped. Refer to Procedure 012-014 (/qs3/pubsys2/xml/en/procedures/20/20-012-014-tr.html).
- Install the fuel pump (if the engine equipped with an air compressor). Refer to Procedure 005-016 (/qs3/pubsys2/xml/en/procedures/20/20-005-016-tr.html).
- Install the lubricating oil filters. Refer to Procedure 007-013 (/qs3/pubsys2/xml/en/procedures/20/20-007-013-

tr.html).

with Electronically Actuated Injector

- Install the ECM cooling plate. Refer to Procedure 006-006 (/qs3/pubsys2/xml/en/procedures/20/20-006-006-tr.html).
- Install the fuel pump. Refer to Procedure 005-016 (/qs3/pubsys2/xml/en/procedures/20/20-005-016-tr.html).

Last Modified: 30-Jun-2006